

ALIAH UNIVERSITY

EVEN (SPRING) SEMESTER EXAMINATION 2021
Compulsory course of Arabic and Islamic Studies
Course code: AI131/AISUGAU01

Max. Marks: 40

Time: 2 hours

Note: Attempt all the questions.

ARABIC

Q1. Answer any ten of the following.

2×10=20

1. How many Sun letters (الحروف الشمسية) have in Arabic alphabet? Write them.
2. Write down the types of pronouns (الضمائر) as you read.
3. Translate the following words. جاهل، انسان، بيت، ثمر.
4. Say whether of the following words are definite or indefinite.
زيد، بلد، الطيب، الفرس، رجل،
5. How many types of words (كلمة) are there in Arabic?
6. Define the nominal sentence with example.
7. What do you mean by nunation (التنوين)?
8. Give four examples of Demonstrative pronoun (أسماء الإشارة).
9. Split the following words. لبن، ماء، ولد، كريم.
10. Join the following letters and make words.
د+ر+س = ر+س+و+ل = ب+ي+ت = ن+ه+ا+ر =
11. Give two examples of possessive (المضاف و المضاف إليه).
12. Write down the definition of اسم (noun) with example.

ISLAMIC STUDIES

Q2. Answer any four of the following.

5×4=20

1. Discuss the Medina life of the Prophet Muhammad (PBUH).
2. What do you know about 'Hadith'? write down.
3. Discuss about Five Daily Prayers.
4. Briefly discuss about 'Zakat' and 'Fasting'.
5. Write a short note about importance of 'social Rights' in Islam.
6. What is the concept of 'Risalah' in Islam?

ALIAH UNIVERSITY

B.Tech End Semester (Odd) Examinations 2022
Subject- Basic Electronics Engineering (ESC ECEUGES01)
MEN, CEN & CSE1st Year 2nd SEM

Full Marks: 80

Time: 3 hrs

(Answer any five)

- Distinguish between insulator, semiconductor and metal on the basis of energy band diagram.
 - Establish the following relation between drift velocity, mobility and electric field for an electron in a semiconductor under thermal equilibrium :- (symbols have usual meaning)
$$v = \mu_n E$$
 - What is Einstein relation? (5+8+3)
- Why thermal stability of *Ge* is lower than *Si* ?
 - Explain the Fermi-Dirac distribution function with graphical interpretation.
 - Show that in n-type semiconductor the Fermi level energy E_F is close to the conduction band edge E_c of the band diagram. (3+8+5)
- Determine the diode dynamic resistance r_d and cut-in voltage V_γ of the forward-bias diode at room temperature with the following parameters :- (symbols have usual meaning)
$$I_S = 1 \text{ nA}, \quad \eta = 2, \quad \text{near the region where the diode current is } 3 \text{ mA.} \quad (8+8)$$
- Show that equilibrium concentrations of electrons (n_0) and holes (p_0) in a semiconductor can be expressed as :- (symbols have usual meaning)
$$(a) \quad n_0 = N_c e^{-\left(\frac{E_c - E_F}{KT}\right)} \quad (b) \quad p_0 = N_v e^{-\left(\frac{E_F - E_v}{KT}\right)}$$
 - Show that the product of electron and hole concentrations under equilibrium is constant and can be expressed as :- $n_0 p_0 = n_i^2$. (symbols have usual meaning) (6+6+4)
- Explain the formation of the depletion region and barrier potential V_0 in a p-n junction diode under equilibrium.
 - Explain the forward bias and reverse bias characteristics curve of p-n junction diode from Shockley diode equation. (4+4+4+4)
- Explain the operation of the clipper circuit with the input shown in Fig. 1, assuming the diode as ideal diode.
 - Draw and explain the output of the clamper circuit shown in Fig. 2 with the given sinusoidal input voltage.
 - Draw the half-wave rectifier out waveform. Find out the amplitude of cosine even harmonic components of it. (4+4+2+6)

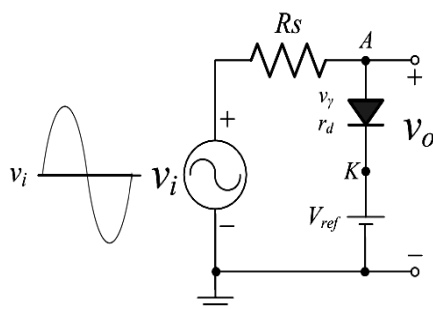


Fig. 1

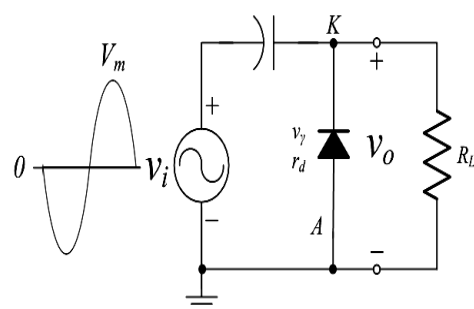


Fig. 2

7. (i) Explain the operation of full wave rectifier circuit using bridge network.
(ii) Show that full-wave rectifier output does not contain sine harmonic components.
(iii) Show that the output DC voltage (V_{dc}) and ripple factor (Γ) of full wave rectifier output is

$$(a) \quad V_{dc} = 0.636 V_m \qquad (b) \quad \Gamma = 48.3 \% \qquad (2+4+3+3+4)$$

8. (i) Explain different current components in a p-n-p transistor in active region with a suitable diagram.
(ii) Establish the relation between current amplification factors α and β .
(iii) Briefly explain the CB, CE and CC mode operation in a transistor. (6+4+6)
9. (i) What is Q-point ? What is need of stability of the Q-point?
(ii) Determine the Q-point values of I_B , I_C and V_{CE} for the circuit given in the Fig-3 . Assume $V_{CC} = 10$ Volt, $R_B = 360 \text{ K}\Omega$ and $R_C = 2 \text{ K}\Omega$.
(iii) Construct the load line and plot the Q points. ((1+3)+6+6)

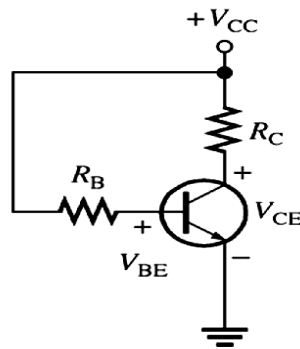


Fig-3

10. (i) Draw the small signal r_e model for CE mode BJT amplifier.
(ii) Also find the following from the small signal analysis of CE amplifier (a) Input impedance
(b) Output impedance (c) Voltage gain (d) Current gain (4+(4×3))

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Autumn Semester Examination - 2021

Subject: Engineering Physics

Subject Code: PHYUGBS01

Full Marks: 80

Time: 3 hrs

- Each question carries 10 marks and answer any eight questions from the following.
1. (a) Find the unit vector which is along the direction of the vector $\vec{A} = \hat{i} + \hat{j} + 2\hat{k}$. (b) A force $\vec{F} = 3\hat{i} - 2\hat{j} + 4\hat{k}$ is acting on a body of mass m and the corresponding position vector is given by $\vec{r} = 2\hat{i} + 3\hat{j} + 5\hat{k}$. Find the torque ($\vec{\tau}$) of the system. (c) Prove that $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b}) = 0$. (d) Define the coefficient of viscosity of a liquid. What is its dimension? (e) Write down Hooke's law. (2+2+2+2+2)
 2. (a) What is a *central force*? (b) Show that a *central force* is conservative. (c) Verify whether the force $\vec{F} = -\frac{\hat{r}}{r^n}$ is conservative or not, where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. (2+3+5)
 3. (a) Define the *centre of mass* (CM) of a system of N -particles. (b) Show that in case of a system of N -particles, the linear momentum ($\vec{P}$), angular momentum ($\vec{L}$) and total energy (E) of the system remain conserved. (1+9)
 4. (a) What do you mean by the nuclear fusion and fission? Give examples. (b) Distinguish between isotopes, isobars and isotones. (c) From the following find out the groups of isotopes and isobars: Ge_{32}^{76} , K_{19}^{39} , C_6^{14} , Ca_{20}^{40} , Ge_{32}^{77} , N_7^{14} . (d) Find X from the following equation: $\text{C}_6^{14} \rightarrow \text{N}_7^{14} + \bar{\nu}_e + X$. (4+2+2+2)
 5. (a) State the de-Broglie's hypothesis and hence calculate the wavelength associated with a body of mass 10 gm moving with a velocity 0.5 m/sec. (b) What is the photo electric effect? Write down the Einstein's photoelectric equation. (c) Write down the time independent Schrödinger equation corresponding to a body of mass m and momentum p , and explain each term. (4+4+2)
 6. (a) State the Coulomb's law of electrostatics. (b) Find the electric field outside a sphere having uniform volume distribution of charge. (c) Determine the amount of charge stored on either plate of a capacitor (4×10^{-6} F) when connected across a 12 V battery. (3+5+2)
 7. (a) What is an electric dipole? (b) Define dipole moment and calculate its SI unit. (c) What is an ideal or point dipole? (d) Derive an expression for the electric field at any point on the equatorial line of an electric dipole. (1+2+2+5)
 8. (a) State the fundamental postulates of kinetic theory of gasses. (b) Calculate the *r.m.s* velocity of hydrogen gas (H_2) at NTP. Given $R = 8.31 \times 10^7$ erg/deg/mol, $M_H = 2$. (c) Find the number of degrees of freedom of a diatomic gas molecule. (d) State the principle of equipartition of energy of gasses. (4+2+2+2)
 9. (a) Define the lattice and basis of a crystal. (b) Determine the relationship between the lattice parameter and the atomic radius r for the monoatomic sc, bcc and fcc structures. (c) Derive the packing fraction of a sc and a bcc lattice. (2+4+4)

“UG” END SEMESTER EXAMINATION – FEBRUARY, 2022
(REGULAR – ODD SEMESTER)
1ST YEAR (1ST SEMESTER) – MECHANICAL ENGINEERING
ENGINEERING MECHANICS
MENUGES01

[Full Marks : 80]

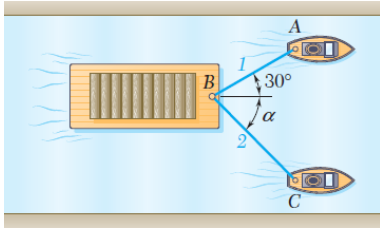
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Assume suitable data(s), if not provided

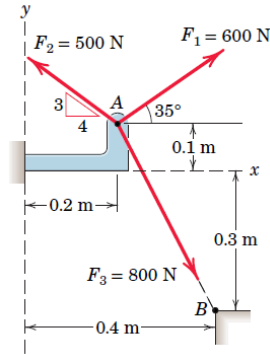
A. Short questions (Answer all five questions):

[5X8 = 40]

1. The forces F_1 , F_2 and F_3 , all of which act on point A of the bracket, are specified in three different ways. Determine each vector into its vector form. (Figure – A1). Also, find the resultant vector, its magnitude and direction with respect to positive x-axis.

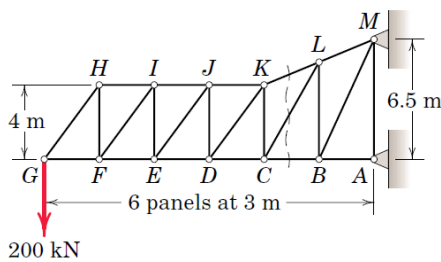


(Figure – A2)

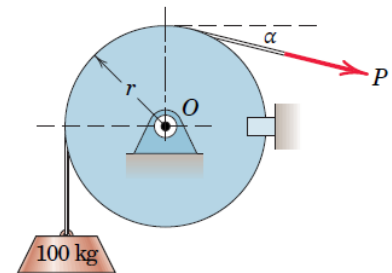


(Figure – A1)

2. Two tugboats are pulling a barge (Figure – A2). If the resultant of the forces exerted by the tugboats is a 5000-kg force directed along the axis of the barge, determine –
 a) The tension in each of the ropes, given that $\alpha = 45^\circ$.
 b) The value of α for which the tension in rope 2 is minimum.

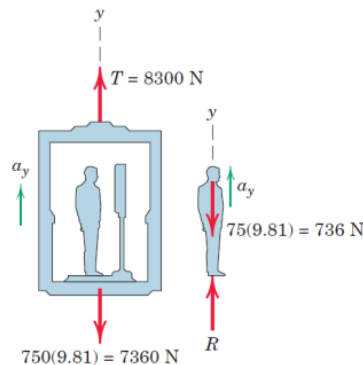


(Figure – A3)



(Figure – A4)

3. Calculate the forces induced in members KL, CL, and CB by the 200 kN load on the cantilever truss. (Figure – A3)
4. A flexible cable which supports the 100-kg load is passed over a fixed circular drum and subjected to a force P to maintain equilibrium. The coefficient of static friction μ between the cable and the fixed drum is 0.30. (Figure – A4)
 a) For $\alpha = 0$, determine the maximum and minimum values which P may have in order not to raise or lower the load.
 b) For $P = 600$ N, determine the minimum value which the angle α may have before the load begins to slip.
5. A 75-kg man stands on a spring scale in an elevator as shown in the figure. During the first 3-sec. of motion from rest, the tension T in the hoisting cable is 8300 N. Find the reading R of the scale in newtons during this interval and the upward velocity v of the elevator at the end of the 3-seconds. The total mass of the elevator, man, and scale is 750-kg. (Figure – A5)



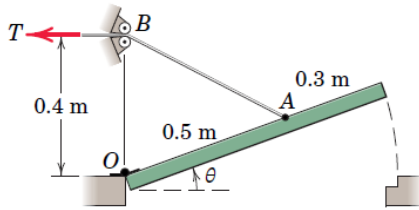
(Figure – A5)

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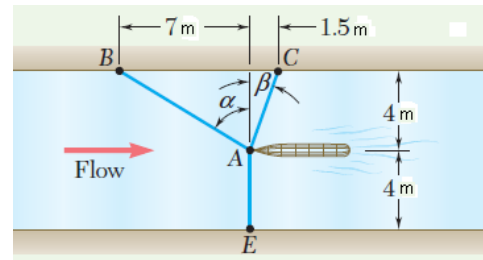
B. Long questions (Answer any four questions):

[4X10 = 40]

1. The trap door OA is raised by the cable AB, which passes over the small frictionless guide pulleys at B. The tension everywhere in the cable is T , and this tension applied at A causes a moment M_O about the hinge at O. Find the maximum and minimum values. What is the physical significance of this ratio? (Figure – B1)

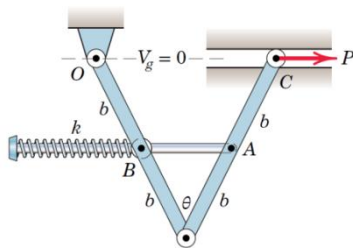


(Figure – B1)

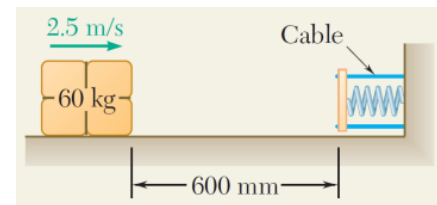


(Figure – B2)

2. For a new sailboat, a designer wants to determine the drag force that may be expected at a given speed. To do so, the designer places a model of the proposed hull in a test channel and uses three cables to keep its bow on the centerline of the channel. Dynamometer readings indicate that for a given speed, the tension is 40 N in cable AB and 60 N in cable AE. Determine the drag force exerted on the hull and the tension in cable AC. (Figure – B2)

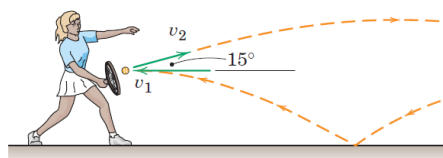


(Figure – B3)



(Figure – B4)

3. The two uniform links, each of mass m , are in the vertical plane and are connected and constrained as shown. As the angle θ between the links increases with the application of the horizontal force P , the light rod, which is connected at A and passes through a pivoted collar at B, compresses the spring of stiffness k . If the spring is uncompressed in the position where $\theta = 0$, determine the force P which will produce equilibrium at the angle θ . (Figure – B3)
4. A spring is used to stop a 60-kg package which is sliding on a horizontal surface. The spring has a constant $k=20\text{-kN/m}$ and is held by cables so that it is initially compressed 120-mm. Knowing that the package has a velocity of 2.5-m/s in the position shown and that the maximum additional deflection of the spring is 40-mm (Figure – B4), determine –
 - a) The coefficient of friction between the package and the surface.
 - b) The velocity of the package as it passes again through the position shown.
5. A tennis player strikes the tennis ball with her racket when the ball is at the uppermost point of its trajectory as shown. The horizontal velocity of the ball just before impact with the racket is $v_1 = 50\text{ m/s}$ and just after impact its velocity is $v_2 = 70\text{ m/s}$ directed at the 15° angle as shown. If the 20 gm ball is in contact with the racket for 0.02 sec, determine the magnitude of the average force $\mathbf{R}$ (in vector form) exerted by the racket on the ball. Also, determine the angle β made by $\mathbf{R}$ with the positive x-axis. (Figure – B5)



(Figure – B5)

===== END =====

ALIAH UNIVERSITY
END SEMESTER EXAMINATION, AUTUMN 2022
PAPER: MATUGBS01 (CALCULUS AND VECTOR ANALYSIS)
Full Marks: 80
Time Allowed: 3 Hours
All symbols have their usual meanings

PART-A.

Answer any **THREE** questions from **PART-A:**

$10 \times 3 = 30$

1. (a) Find out the domain of definition of the real function $f(x) = \frac{1}{\sqrt{|x|-x}}$. [1]
 (b) Define monotonically increasing function. Prove that $f : [0, \infty) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x+1}{3x+5}$ is a strictly monotonically increasing function. [4]
 (c) Show that $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x}\right)^{\frac{1}{x^2}} = e^{-\frac{1}{6}}$. [5]
2. (a) define continuous function. Using $\epsilon - \delta$ definition show that the function $f(x) = \cos x$ is continuous for all real values of x . [5]
 (b) State Leibnitz's theorem. Using Leibnitz's theorem show that $(1-x^2)y_{n+2} - (2n+3)xy_{n+1} - (n-1)^2y_n = 0$ where $y = \frac{\sin x}{\sqrt{1-x^2}}$ with $|x| < 1$. [5]
3. (a) State Rolle's theorem. verify Rolle's theorem for the polynomial $f(x) = 2x^3 - x^2 - 6x + 3$. [5]
 (b) Let $f(x) = \frac{1}{x^2}$ and $g(x) = \frac{1}{x}$, for all $x \in [a, b]$ with $0 < a < b$. Then by Cauchy's mean value theorem, there exists a $c \in (a, b)$ such that $\frac{f'(c)}{g'(c)} = \frac{f(b)-f(a)}{g(b)-g(a)}$. Show that c is harmonic mean of a and b . [5]
4. (a) Define directional derivative in $\mathbb{R}^2$. Show that the directional derivative of the function $f(x, y) = xy$ at (a_1, a_2) in the direction $(1, 2)$ is $a_2 + 2a_1$. [3]
 (b) Let

$$f(x, y) = \begin{cases} \frac{x^2y}{x^4+y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Show that $f(x, y)$ has a directional derivative at $(0, 0)$ in any direction (v_1, v_2) but $f(x, y)$ is not continuous at $(0, 0)$. [4]

- (c) Let

$$f(x, y) = \begin{cases} \frac{xy}{x^2+y^2} & \text{where } x^2 + y^2 \neq 0 \\ 0 & \text{where } x^2 + y^2 = 0 \end{cases}$$

Show that $f_x(0, 0)$ and $f_y(0, 0)$ exist; but the directional derivative does not exist at $(0, 0)$ in the direction $(1, 1)$. [3]

PART-B

Answer any **TWO** questions from **PART-B**:

$10 \times 2 = 20$

5. (a) Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1 + \sin x + \cos x} dx$. [5]
(b) Evaluate $\int_0^{\frac{\pi}{2}} \frac{x}{\sin x + \cos x} dx$. [5]
6. (a) Assuming convergency evaluate, $\int_0^{\frac{\pi}{2}} \log(\sin x) dx$. [5]
(b) Examine the convergence of $\int_1^2 \frac{x}{\sqrt{2-x}} dx$ and evaluate if possible. [5]
7. (a) Show that $\int_a^b (x-a)^3(b-x)^2 dx = \frac{(b-a)^6}{60}$. [5]
(b) If m and n are positive integers then show that
- $$B(n, m) = \frac{(m-1)!(n-1)!}{(m+n-1)!}$$
- [3]
- (c) Find, if possible, the value of $B(\frac{5}{2}, 6)$. [2]

PART-C.

Answer any **THREE** questions from **PART-C**:

$10 \times 3 = 30$

8. (a) Explain 'Gibbs phenomenon' with an example. [2 + 1]
(b) Find the Fourier Trigonometric series of the function,

$$f(x) = \begin{cases} \cos 2x & \text{when } 0 < x < \frac{\pi}{4} \\ 0 & \text{when } \frac{\pi}{4} < x < \frac{\pi}{2} \end{cases}$$

use the expansion to show $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{4n^2-1} = \frac{\pi-2}{4}$. [5 + 2]

9. (a) Find the Fourier Trigonometric series of the function,

$$f(x) = x + x^2 \quad \text{when } -\pi < x < \pi$$

use the expansion to show $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$. [4 + 1]

- (b) Verify Gauss divergence theorem for $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ taken over the rectangular parallelepiped bounded by $x = -a$, $y = -b$, $z = -c$, $x = a$, $y = b$, $z = 2c$. [5]

10. (a) Prove that $\text{curl} \vec{A} \times \vec{B} = (\vec{B} \cdot \vec{\nabla}) \vec{A} - \vec{B}(\vec{\nabla} \cdot \vec{A}) - (\vec{A} \cdot \vec{\nabla}) \vec{B} + \vec{A}(\vec{\nabla} \cdot \vec{B})$ [6]
 (b) Find the directional derivative of the function

$$f(x, y, z) = x^2 - y^2 + 2z^2$$

at $P(1, 2, 3)$ in the direction of vector joining $R(3, 4, 5)$ and $S(-1, 3, 7)$.

What is the greatest rate of increase of $f(x, y, z)$ at R ? [3 + 1]

11. (a) Verify Stokes's theorem for the hemisphere $x^2 + y^2 + z^2 = 9$, $z \geq 0$ and the vector field $\vec{F} = (z^2 - y)\hat{i} + (x - 2yz)\hat{j} + (2xz - y^2)\hat{k}$. [5]
 (b) State Green's theorem. Verify Green's theorem in the plane of xy for $\vec{F} = (3x^2 - 8y^2)\hat{i} + (2y - 3xy)\hat{j}$ where C is boundary of region bounded by $x = a$, $y = 0$ and $x + y = a$. [1 + 4]